

ApexMaths — Proof of AWS Database Usage

Amazon Aurora PostgreSQL (Serverless v2) · RDS Data API · eu-west-2 (London)

Evidence flow: Vercel → IAM (OIDC, no static keys) → Aurora PostgreSQL

Aurora75's proj...

Hobby

vercel-v0-and-aws-hackathon

Overview

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Overview

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Speed Insights

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Open In

Repository

Instant Rollback

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6 topics

Deployment

vercel-v0-and-aws-hackathon-k8ursvj3x-aurora75-s-projects.vercel.app

Domains

vercel-v0-and-aws-hackathon.vercel.app

Status

Created

Ready

2d ago by mirkovicUK

Source

main

238a948 Merge pull request #61 from mirkovicUK/dev

Deployment Settings

5 Recommendations

To update your Production Deployment, push to the main branch.

Deployments

Production Checklist 4/5

Connect Git Repository

Add Custom Domain

Preview Deployment

Enable Web Analytics

Enable Speed Insights

Observability 6h

Edge Requests

11

Function Invocations

9

Error Rate

0%

Analytics 1w

20 Visitors

33%

0 online

Active Branches

Search...

Status 6/7

Aurora75

https://vercel-v0-and-aws-hackathon.vercel.app

OIDC federation enabled — Vercel obtains short-lived AWS credentials at runtime. No static AWS keys anywhere.

Aurora75's proj...

Hobby

vercel-v0-and-aws-hackathon

Project Settings

Security

Find...

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Settings

General

Build and Deployment

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Project Members

Drains

Security

Networking

Advanced

Learn more about [Git Fork Protection](#)

Save

Secure Backend Access with OIDC Federation

Authenticate with your cloud providers using OpenID Connect JSON Web Tokens.

Issuer Mode

Team

Recommended

https://oidc.vercel.com/aurora75-s-projects

Global

https://oidc.vercel.com

Claim	Kind	Value
iss	Issuer	https://oidc.vercel.com/aurora75-s-projects
aud	Audience	https://vercel.com/aurora75-s-projects
sub	Subject	owner:aurora75-s-projects:project:vercel-v0-and-aws-hackathon:environment:production
scope	Scope	owner:aurora75-s-projects:project:vercel-v0-and-aws-hackathon:environment:production
iat	Issued At	1782654969
nbf	Not Before	1782654969
exp	Expiration	1782658569

Learn more about [Secure Backend Access with OIDC Federation](#)

Save

Deployment Retention Policy

Define a retention period for all Deployments for this Project. Deployments will be deleted after the defined time.

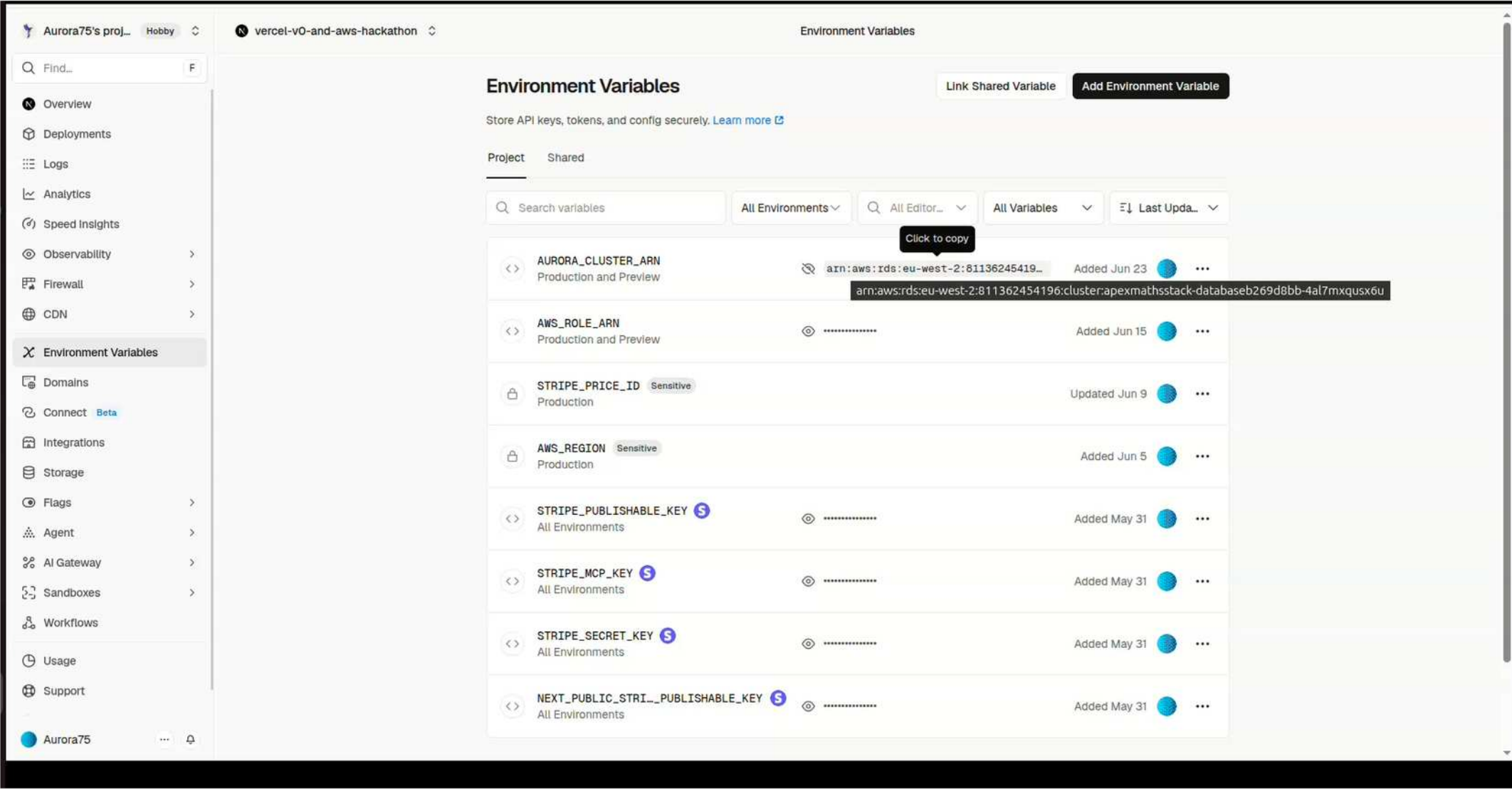
Canceled Deployments

ApexMaths · Proof of AWS Database Usage

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VERCEL — environment configuration

Vercel env vars reference the database by ARN (AURORA_CLUSTER_ARN) — the app targets Aurora by ARN, not a password.



IAM – trust relationship (no static keys)

The IAM role is assumable only by Vercel deployments via the OIDC issuer — short-lived STS credentials, no long-lived keys.

IAM > Roles > apexmaths-vercel

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

- Roles
- Policies
- IAM users
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
 - Resource analysis
 - Unused access
 - Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies

Related consoles

- IAM Identity Center

apexmaths-vercel Info

Assumed by Vercel deployments via OIDC federation (no static keys).

Delete

Edit

Summary

Creation date
June 15, 2026, 12:39 (UTC+01:00)

Last activity

✓ 16 hours ago

ARN
arn:aws:iam::811362454196:role/apexmaths-vercel

Maximum session duration
1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Trusted entities

Edit trust policy

Entities that can assume this role under specified conditions.

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Principal": {  
7         "Federated": "arn:aws:iam::811362454196:oidc-provider/oidc.vercel.com/aurora75-s-projects"  
8       },  
9       "Action": "sts:AssumeRoleWithWebIdentity",  
10      "Condition": {  
11        "StringEquals": {  
12          "oidc.vercel.com/aurora75-s-projects:aud": "https://vercel.com/aurora75-s-projects"  
13        },  
14        "StringLike": {  
15          "oidc.vercel.com/aurora75-s-projects:sub": [  
16            "owner:aurora75-s-projects:project:*:environment:production",  
17            "owner:aurora75-s-projects:project:*:environment:preview"  
18          ]  
19        }  
20      }  
21    }  
22  ]  
23 }
```

CloudShell Feedback Console Mobile App

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IAM — least-privilege permissions

rds-data:* actions scoped to this one Aurora cluster (DML only) — the app's access to Aurora is via the RDS Data API.

The screenshot shows the AWS IAM console interface for the role 'apexmaths-vercel'. The left sidebar contains navigation links for Identity and Access Management (IAM), Access Management, and Access reports. The main content area displays the role's summary, including its creation date (June 15, 2026, 12:39 UTC+01:00) and last activity (16 hours ago). The 'Permissions policies' tab is selected, showing a list of attached policies. One policy, 'VercelRoleDefaultPolicy77548288', is listed with a 'Customer inline' type and 0 attached entities. The policy's JSON is expanded, showing a single statement that grants 'Allow' permissions for 'rds-data' actions (BatchExecuteStatement, BeginTransaction, CommitTransaction, ExecuteStatement, RollbackTransaction) on the resource 'arn:aws:rds:eu-west-2:811362454196:cluster:apexmathsstack-databaseb269d8bb-4a17mxqusx6u'. The policy is associated with the 'AuroraDataApi' SID.

Summary

Creation date
June 15, 2026, 12:39 (UTC+01:00)

Last activity
16 hours ago

ARN
arn:aws:iam::811362454196:role/apexmaths-vercel

Maximum session duration
1 hour

Permissions policies (1)

You can attach up to 10 managed policies.

Filter by Type
All types

Policy name	Type	Attached entities
VercelRoleDefaultPolicy77548288	Customer inline	0

VercelRoleDefaultPolicy77548288

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Action": [
6         "rds-data:BatchExecuteStatement",
7         "rds-data:BeginTransaction",
8         "rds-data:CommitTransaction",
9         "rds-data:ExecuteStatement",
10        "rds-data:RollbackTransaction"
11      ],
12      "Resource": "arn:aws:rds:eu-west-2:811362454196:cluster:apexmathsstack-databaseb269d8bb-4a17mxqusx6u",
13      "Effect": "Allow",
14      "Sid": "AuroraDataApi"
15    }
16  ]
17 }
```


AURORA POSTGRESQL — end-to-end ARN match

The Aurora cluster ARN matches the Vercel env var and the IAM policy — confirming the Vercel -> IAM -> Aurora link.

Aurora and RDS

Dashboard

Databases

Query editor

Performance insights

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Exports in Amazon S3

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Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

Recommendations 4

Certificate update

apexmathsstack-databaseb269d8bb-4al7mxqusx6u

Related

Filter by databases

DB identifier	Status	Role	Engine	Region & AZ	Size	Recommendations
apexmathsstack-databaseb269d8bb-4al7mxqusx6u	Available	Regional C...	Aurora Po...	eu-west-2	1 Instance	3 Informational and 2
apexmathsstack-databasewriter2462cc03-ygjqad1sjtf3	Available	Writer Ins...	Aurora Po...	eu-west-2b	Aurora serverless (0.5 - 2 A...	2 Informational

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Tags

Recommendations

Database

Configuration

DB cluster role
Regional cluster

Engine version
16.11

RDS Extended Support
Enabled

Resource ID
cluster-ETVSWR6Q2UKQX5DDK2YYMZUYEQ

Cluster storage configuration
Aurora Standard

Amazon Resource Name (ARN)
arn:aws:rds:eu-west-2:811362454196:cluster:apexmathsstack-databaseb269d8bb-4al7mxqusx6u

Network type

Capacity type
Provisioned

Local read replica write forwarding
Disabled

DB cluster ID
apexmathsstack-databaseb269d8bb-4al7mxqusx6u

DB cluster parameter group
default:aurora-postgresql16

Deletion protection
Disabled

Limitless Database
Disabled

Authentication

IAM DB authentication
Not enabled

Kerberos authentication
Not enabled

Master username
apexadmin

Master password

Availability

Multi-AZ
No

Encryption

Encryption key
AWS managed KMS key
aws/rds

Monitoring

Monitoring type
Database Insights - Standard

Performance Insights
Disabled

Enhanced Monitoring
Disabled

CloudShell

Feedback

Console Mobile App

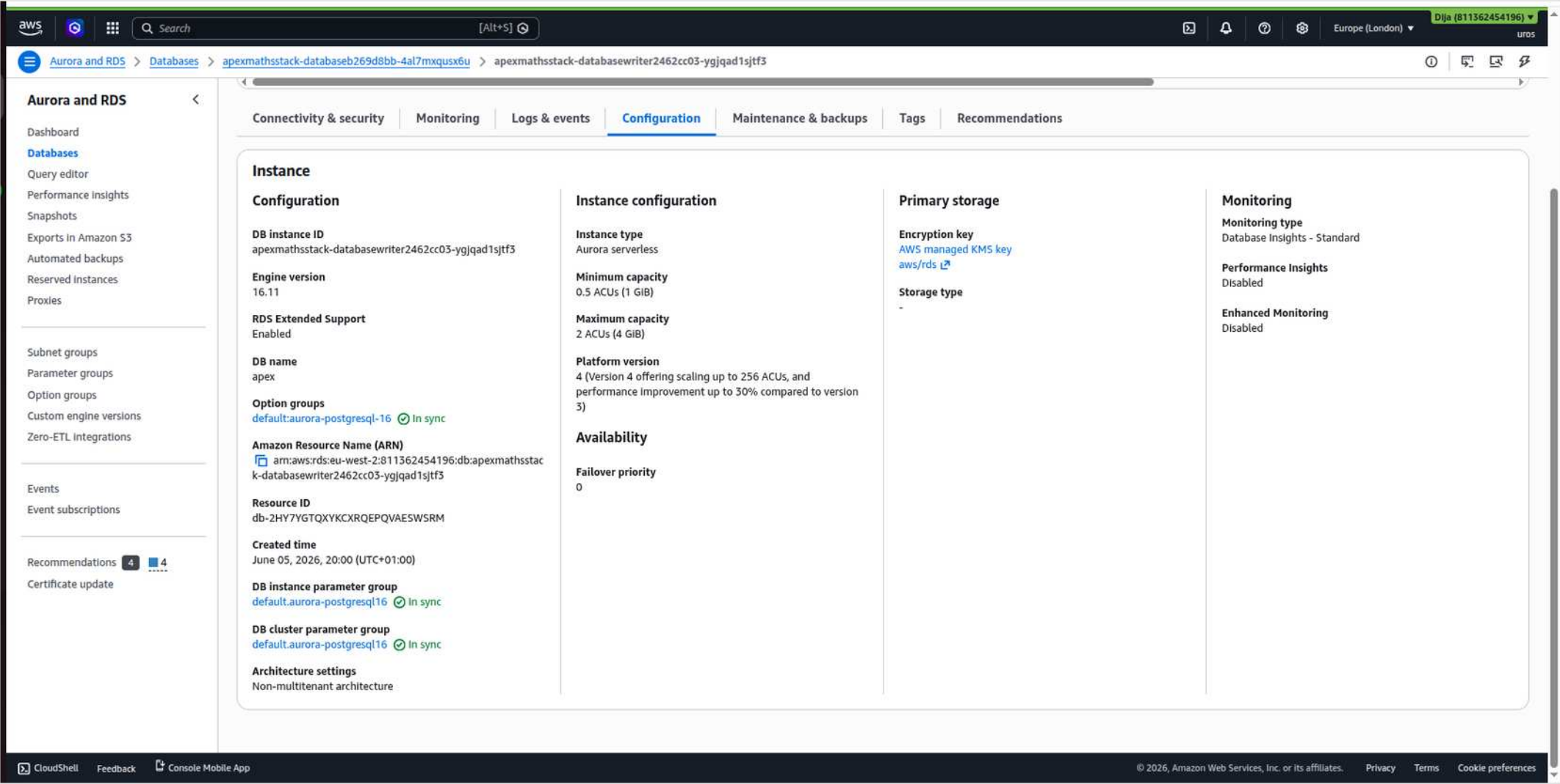
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Amazon Aurora PostgreSQL, Serverless v2, region eu-west-2 (London).



AURORA POSTGRESQL — cluster configuration (2/2)

Single Serverless v2 instance (deliberate cost choice); Aurora storage is replicated across 3 Availability Zones.



AURORA POSTGRESQL – Data API enabled

describe-db-clusters -> HttpEndpointEnabled = true. The RDS Data API is the cluster's HTTP endpoint the app calls.

Aurora and RDS

Databases

apexmathsstack-databaseb269d8bb-4al7mxqusx6u > apexmathsstack-databasewriter2462cc03-ygjqad1sjtf3

Aurora and RDS

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apexmathsstack-databasewriter2462cc03-ygjqad1sjtf3

Related

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DB identifier	Status	Role	Engine	Region & AZ	Size	Recommendations
apexmathsstack-databaseb269d8bb-4al7mxqusx6u	Available	Regional C...	Aurora Po...	eu-west-2	1 Instance	3 Informational and 2
apexmathsstack-databasewriter2462cc03-ygjqad1sjtf3	Available	Writer Ins...	Aurora Po...	eu-west-2b	Aurora serverless (0.5 - 2 A...	2 Informational

Connectivity & security

Monitoring

Logs & events

Configuration

Maintenance & backups

Tags

Recommendations

CloudShell

eu-west-2 +

```
- $ aws rds describe-db-clusters \
> --db-cluster-identifier apexmathsstack-databaseb269d8bb-4al7mxqusx6u \
> --query "DBClusters[0].HttpEndpointEnabled"
true
~ $
```

CloudShell

Feedback

Console Mobile App

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AURORA POSTGRESQL – live query via Data API

RDS Query Editor (runs on the Data API): `SELECT count(*) FROM questions` — real seeded data in the question bank.

AWS

[Alt+S]

Search

Europe (London)

Dija (811362454196)

uros

Aurora and RDS

> Query editor: apexmathsstack-databaseb269d8bb-4al7mxqusx6u

Aurora and RDS

<

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Certificate update

Connected to database apexmathsstack-databaseb269d8bb-4al7mxqusx6u successfully.

Editor

Recent

Saved queries

Query editor: apexmathsstack-databaseb269d8bb-4al7mxqusx6u

Query 1 X +

1 SELECT count(*) FROM questions;

2

Run

Save

Clear

Amazon Q profile 2:1 pgSQL

Output

Result set 1 (1)

Rows returned (1)

Search rows

count

643

Export to csv

< 1 >

AURORA POSTGRESQL — integrity in-engine

A partial UNIQUE index enforces one active session per child (uniq_active_session_per_child WHERE status='active').

AWS

Search

[Alt+S]

Europe (London)

Dija (811362454196)

uros

Aurora and RDS

> Query editor: apexmathsstack-databaseb269d8bb-4al7mxqusx6u

Aurora and RDS

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Certificate update

Connected to database apexmathsstack-databaseb269d8bb-4al7mxqusx6u successfully.

Editor

Recent

Saved queries

Query editor: apexmathsstack-databaseb269d8bb-4al7mxqusx6u

Query 1 X +

1 SELECT indexname, indexdef

2 FROM pg_indexes

3 WHERE tablename = 'sessions'

4 AND indexname = 'uniq_active_session_per_child';

5

6

Amazon Q profile 5:1 pgSQL

Run

Save

Clear

Output

Result set 1 (1)

Rows returned (1)

Search rows

< 1 >

indexname

indexdef

uniq_active_session_per_child

CREATE UNIQUE INDEX uniq_active_session_per_child ON public.sessions USING btree (child_id) WHERE (status = 'active')::session_status

Export to csv

AURORA POSTGRESQL — relational data model

Foreign keys with CASCADE / SET NULL / NO ACTION — GDPR erasure is a single cascading DELETE FROM parents.

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Certificate update

Query editor: apexmathsstack-databaseb269d8bb-4a7mxqusx6u

SELECT tc.table_name,
tc.constraint_name,
rc.delete_rule
FROM information_schema.table_constraints tc
JOIN information_schema.referential_constraints rc
ON tc.constraint_name = rc.constraint_name
WHERE tc.constraint_type = 'FOREIGN KEY'
ORDER BY rc.delete_rule, tc.table_name;

Run Save Clear

Output

Result set 1 (10)

Rows returned (10)

Search rows

table_name	constraint_name	delete_rule
children	children_parent_id_fkey	CASCADE
progress	progress_child_id_fkey	CASCADE
review_reports	review_reports_session_id_fkey	CASCADE
session_answers	session_answers_session_id_fkey	CASCADE
sessions	sessions_child_id_fkey	CASCADE
sessions	sessions_parent_id_fkey	CASCADE
subscriptions	subscriptions_parent_id_fkey	CASCADE
session_answers	session_answers_question_id_fkey	NO ACTION
contact_messages	contact_messages_parent_id_fkey	SET NULL
revenue_events	revenue_events_parent_id_fkey	SET NULL

Export to csv

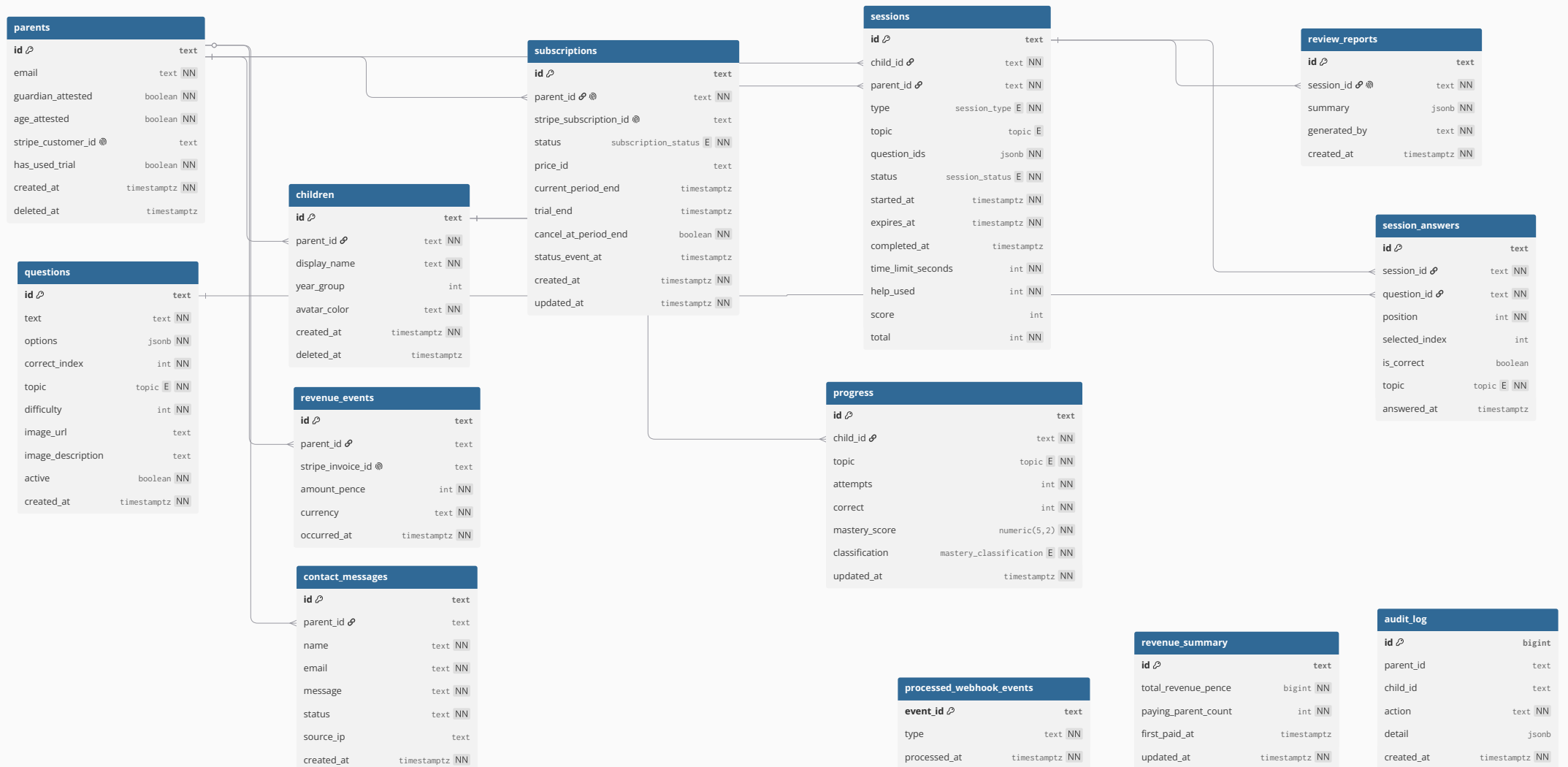
CloudShell Feedback Console Mobile App

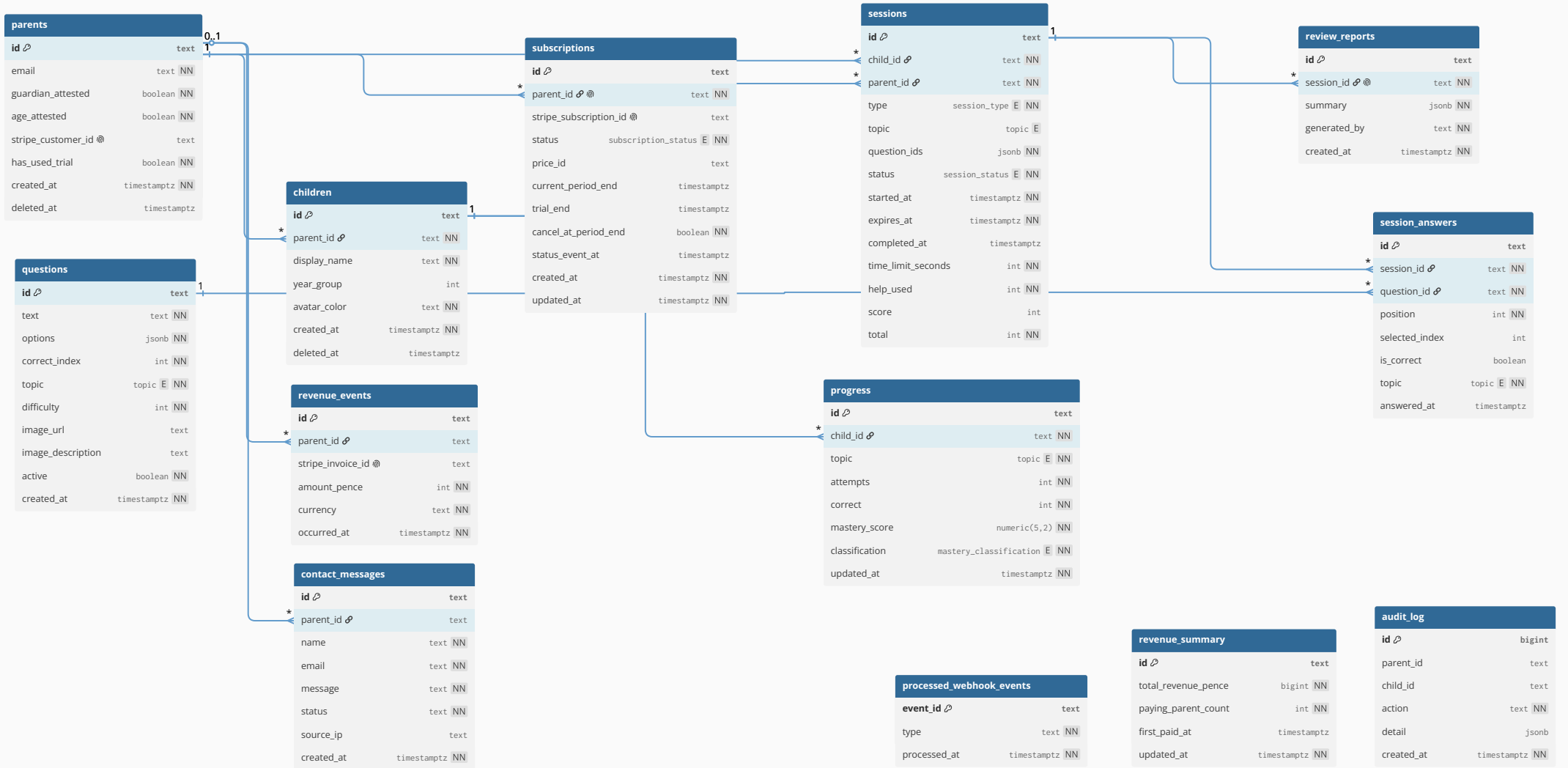
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Data Model — Entity-Relationship Diagram

Generated from the live Aurora PostgreSQL schema (dbdiagram.io). The screenshots prove the model is real and queryable; the next pages show the relational design visually.

- **Page 1** — Full schema: every table and column.
- **Page 2** — Relationships: foreign-key connections between tables (highlighted).
- **Page 3** — Enumerated types: topic, session_type, session_status, subscription_status, mastery_classification.





```
Enum topic {  
    number  
    fractions_decimals_percentages  
    ratio_proportion  
    algebra  
    geometry  
    data_handling  
}
```

```
Enum session_type {  
    warmup  
    topic  
    mock  
    adaptive  
}
```

```
Enum session_status {  
    active  
    completed  
    expired  
    abandoned  
}
```

```
Enum subscription_status {  
    trialing  
    active  
    past_due  
    canceled  
    incomplete  
    unpaid  
}
```

```
Enum mastery_classification {  
    needs_focus  
    developing  
    strong  
    insufficient_data  
}
```